

Local scalar-transfer rigidity at strictly constrained endpoints

Research note; independently reviewed

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This extends the regular interior-boundary result in `../independent_directions/scalar_transfer_boundary.tex`. The statement explicitly assumes strict endpoint signs. It makes no claim about endpoint strata where the normal derivative vanishes.

Theorem 1. *Fix $m \geq 2$ and a scalar array $a : [K] \rightarrow [0, 1]$, nondecreasing, with at most $m - 1$ distinct levels in $(0, 1)$. Give its rows fixed positive probability weights. Endpoint levels 0 and 1 may also occur. Suppose a degree- m polynomial*

$$P(t) = \sigma t^m + \sum_{s=0}^{m-1} \gamma_s t^s, \quad \sigma \in \{-1, 1\},$$

has the following properties:

- $P'(a_l) = 0$ at every interior level;
- $P''(a_l) > 0$ at every interior level containing at least two rows;
- $P'(0) > 0$ if the level 0 occurs, and $P'(1) < 0$ if level 1 occurs.

Then, in a neighborhood of the scalar array with all columns equal to a , every nondecreasing $[0, 1]$ -valued array $x_1, \dots, x_m$ with exchangeable mixed moments

$$\mathbb{E} \prod_{j \in S} x_j = \mu_{|S|}$$

and with $\mu_s = \mathbb{E} a^s$ for $1 \leq s < m$ satisfies

$$\sigma(\mu_m - \mathbb{E} a^m) \geq 0.$$

In the equality case all endpoint blocks are unchanged; on every interior block all columns have mean a_l , and at most one column is nonconstant. Thus all mixed moments through degree m equal those of the original scalar array.

Proof. All constants depend only on the fixed scalar array and m . Let $\delta = \max_{j,q} |x_j(q) - a(q)|$. On each interior level block write

$$x_j = a_l + b_{jl} + e_j, \quad \mathbb{E}_l e_j = 0,$$

where $\mathbb{E}_l$ is the conditional weighted mean on that block. Put

$$C_{lij} = \mathbb{E}_l(e_i e_j) \geq 0, \quad S = \sum_{l \text{ interior}} \sum_{i < j} C_{lij}, \quad B = \max_{j,l \text{ interior}} |b_{jl}|.$$

If there are no interior levels, take $B = S = 0$. For endpoint blocks put $b_{j0} = \mathbb{E}_0 x_j \geq 0$ and $b_{j1} = \mathbb{E}_1(x_j - 1) \leq 0$, omitting absent blocks, and set

$$T = \sum_j (b_{j0} - b_{j1}) \geq 0.$$

We use the elementary inequality $\mathbb{E}_l |fg| \leq 3\mathbb{E}_l(fg)$ for two nondecreasing mean-zero functions. Indeed, represent them as positive combinations of centered steps $h_s(u) = \mathbf{1}_{u \geq s} - (1 - s)$ after embedding the weighted block in $[0, 1]$. For $s \leq t$,

$$\mathbb{E} h_s h_t = s(1 - t), \quad \mathbb{E} |h_s h_t| = s(1 - t)[1 + 2(t - s)] \leq 3s(1 - t).$$

Consequently, any product of at least two distinct interior errors has absolute mean at most $3(2\delta)^{s-2} C_{lij}$, where s is its number of factors and i, j are any two of them.

The lower mixed moments of the block-constant interior means differ from their prescribed scalar values by $O(S + T)$. Here interior fluctuations contribute $O(S)$, while endpoint blocks contribute $O(T)$ because their deviations have one fixed sign and are bounded by δ . The derivative of the lower-moment map in the interior variables b_{jl} is injective. A kernel vector would satisfy, for every s -element column set I , $1 \leq s < m$,

$$\sum_{j \in I} \sum_{l \text{ interior}} w_l a_l^{s-1} b_{jl} = 0,$$

where w_l is the fixed block weight. Comparing sets differing in one index makes each inner sum zero. The first r equations, where r is the number of interior levels, form an invertible weighted Vandermonde system for each

column. A bounded left inverse and Taylor expansion therefore give $B \leq C(S + T + B^2)$, hence

$$B = O(S + T)$$

after shrinking the neighborhood.

The symmetric multiaffine polarization of P at a row $(a_l + d_1, \dots, a_l + d_m)$ is

$$P(a_l) + \sum_{s=1}^m \frac{P^{(s)}(a_l)}{s!} \frac{e_s(d_1, \dots, d_m)}{\binom{m}{s}}.$$

On interior blocks the linear term vanishes. Their quadratic main term is

$$\sum_{l \text{ interior}} w_l \frac{P''(a_l)}{m(m-1)} \sum_{i < j} C_{lij}.$$

Singleton blocks contribute no covariance. Thus this term is at least cS if S can be nonzero; all errors, including products of block mean shifts, are $O(\delta S + (S + T)^2)$.

On an endpoint block, every deviation $d_j = x_j - a_l$ has the same sign. A product of $s \geq 2$ of them has absolute mean at most $\delta^{s-1} \mathbb{E}_l |d_j|$ for any chosen factor. Hence the endpoint contribution is its linear term plus $O(\delta T)$. By the strict endpoint signs and the positive block weights, its linear term is at least cT . Combining the two types of blocks gives

$$\mathbb{E} \text{Pol}(P)(x_1, \dots, x_m) - \mathbb{E}P(a) \geq c(S + T) - C(\delta(S + T) + (S + T)^2) \geq 0$$

for sufficiently small δ , with strict inequality if $S + T > 0$. Exchangeability and the prescribed lower moments identify the left side as $\sigma(\mu_m - \mathbb{E}a^m)$.

Equality forces $S = T = 0$, hence $B = 0$. The endpoint blocks are fixed because their mean one-sided deviations vanish. On an interior block, zero covariance between two monotone centered functions forces one to be constant: the monotone covariance formula has a strictly positive first-last term if both vary. Thus at most one column varies on that block, and its mean is the scalar level. Every distinct-coordinate product has the scalar block mean, proving the equality statement. $\square$

At a scalar constrained local minimum, the usual one-sided endpoint conditions give weak signs $P'(0) \geq 0$ and $P'(1) \leq 0$. The theorem covers the strictly complementary case. Vanishing endpoint derivatives, singular interior strata, and global components remain outside its scope. In particular this note does not establish general scalar transfer or a universal quadratic per-die lower bound.